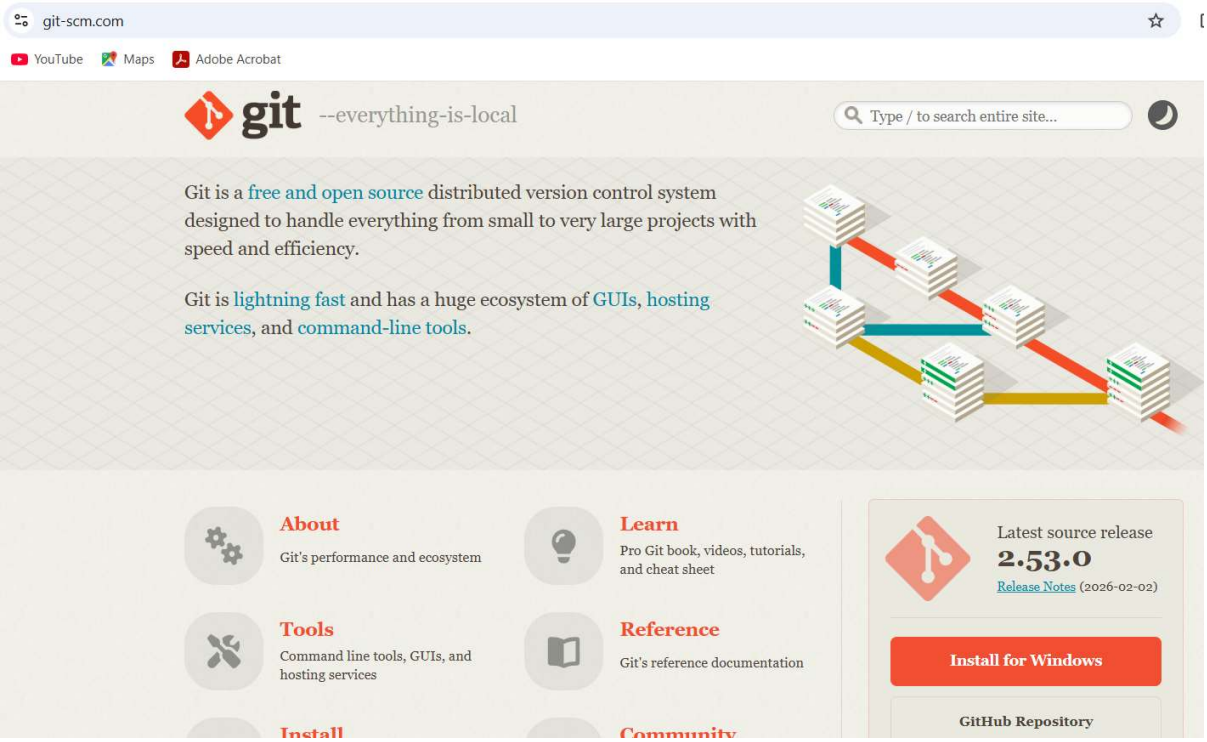
GIT FOR BEGINEERS

What is github?

GitHub is a cloud-based platform that acts as a digital hub for developers to store, manage, share, and collaborate on code projects. It uses Git, a version control system, allowing teams to track changes, work simultaneously on code, and merge updates safely.

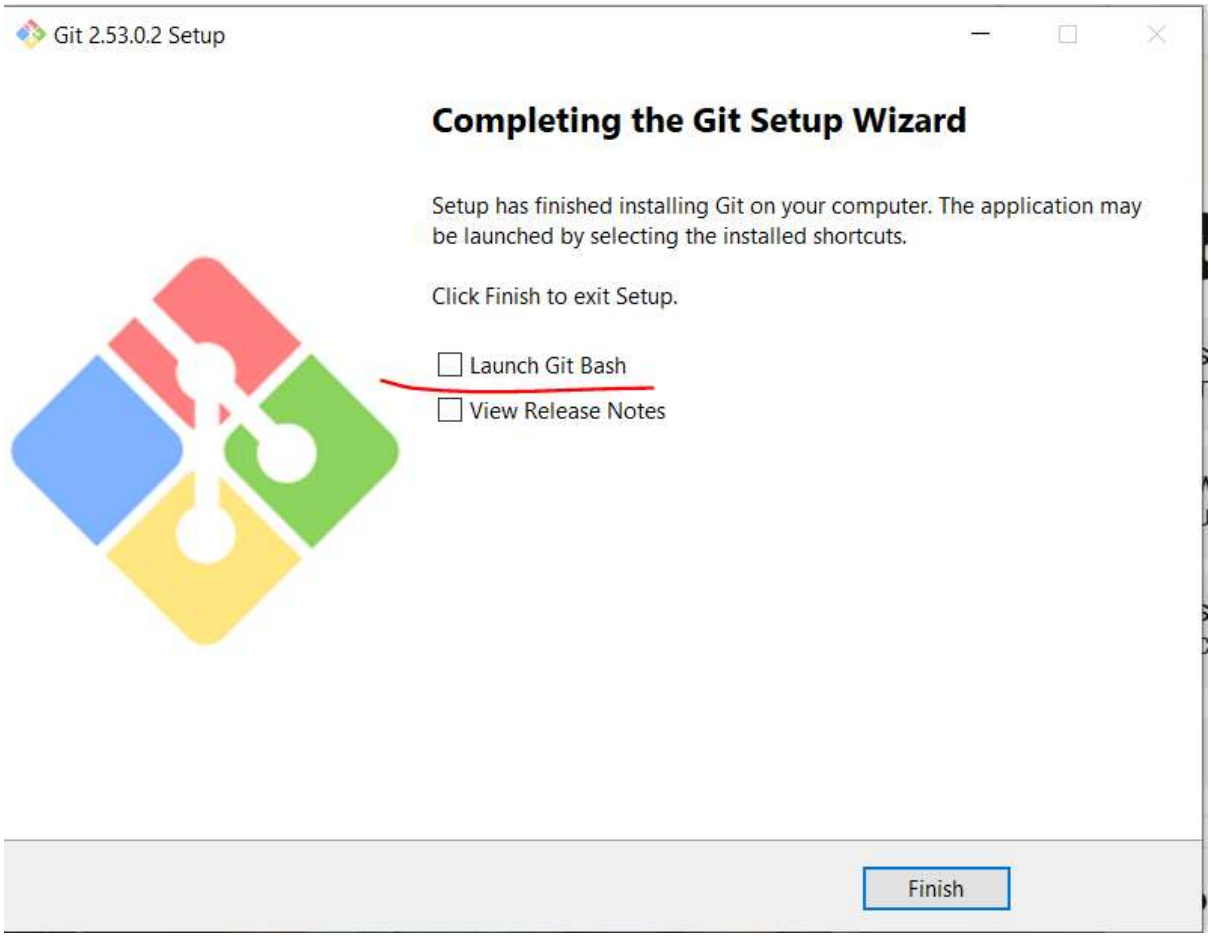
- Version control system
- Easily recover files
- Who introduced issue and when?
- Rollback to previously working state.
- Free of cost

Git Downloads for windows



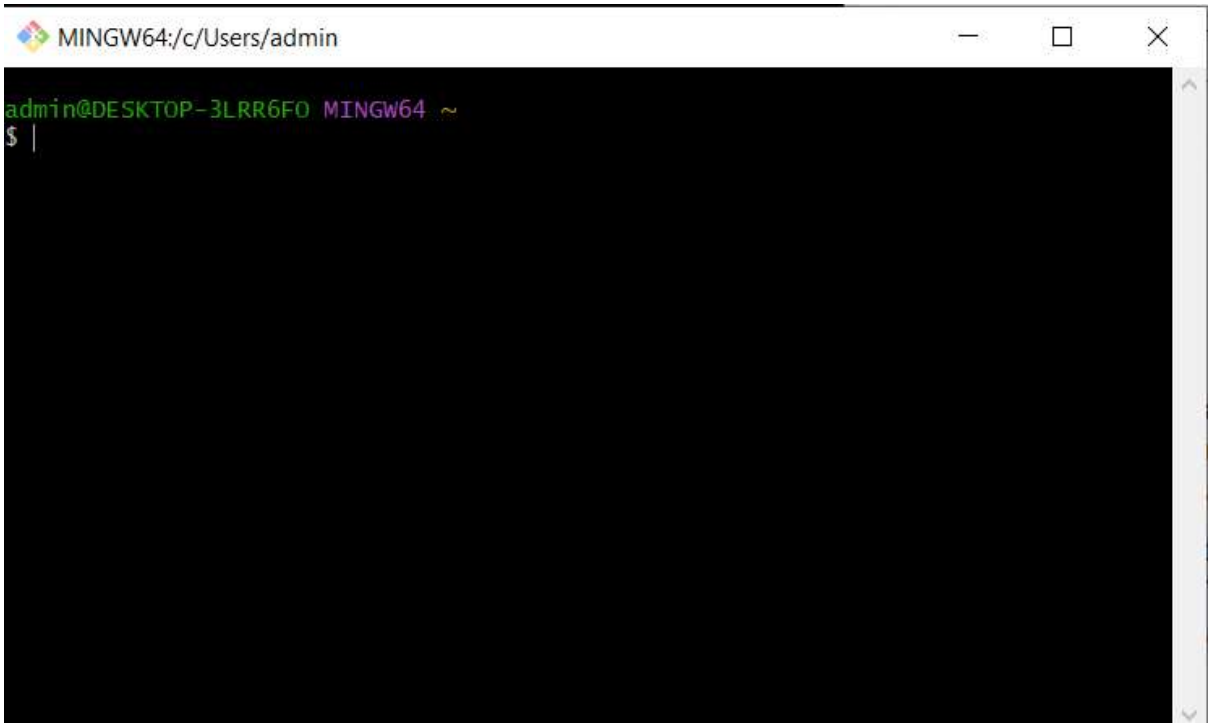
Open the .exe file and install it.

Last screen



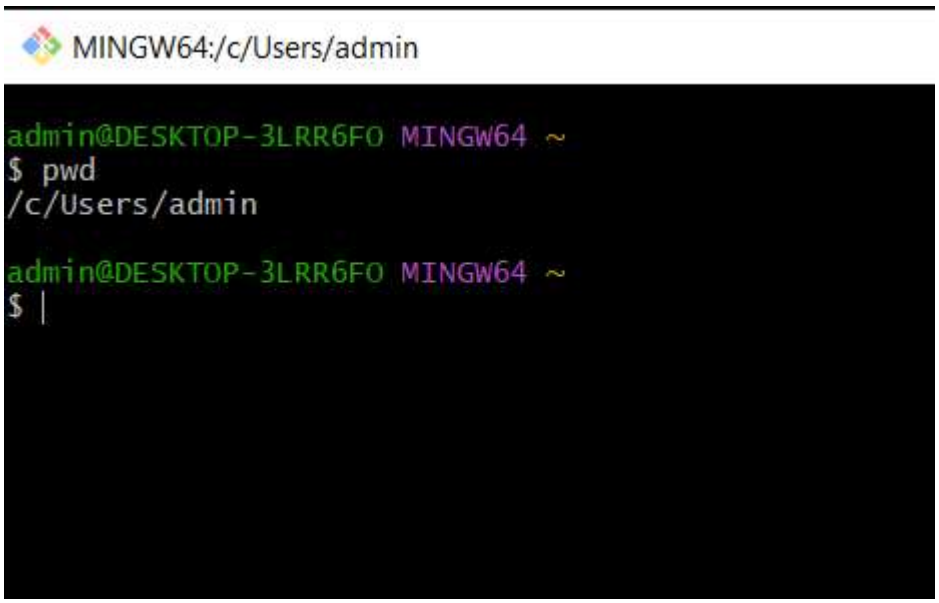
Either click “launch Git Bash” and Finish OR only Finish

I clicked on “launch Git Bash” and Finish



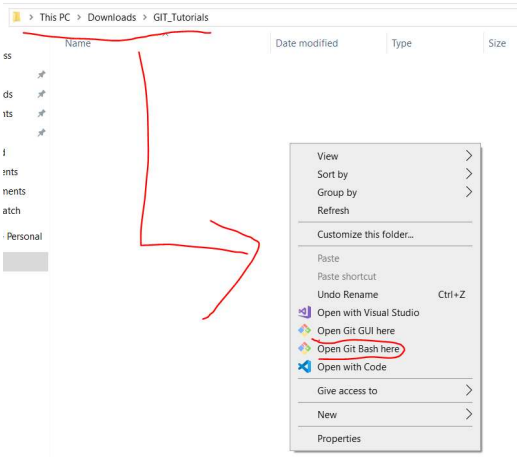
It is command line tool similar like command prompt.

- **To check present working directory**
  Command → pwd



⇨ Open git bash for any folder

- **Create a folder**
- **Go to inside folder**
- **Right click and click on git bash**



Or

- **Search in window → git bash**
- **Click on it**
- **Once it will open, with help of cd , reach to folder.**

⇨ Configure your Name & Email

git config –global user.name "Prateek"

git config –global user.email "Prateek.123@gmail.com"

git config –list

- **Created folder and also few different files.**

- **Check git status**

  Command → git status

```
admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/GIT_Tutorials (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        New Microsoft Word Document.docx
        pp.txt.txt

nothing added to commit but untracked files present (use "git add" to track)

admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/GIT_Tutorials (master)
$
```

- **Initialized git**

  Command → git init

- **git add –a**

- **git log**

- **How to check modified code**

  git status

  if available, it shows in red

- **Do, git add ***

  It will go green.

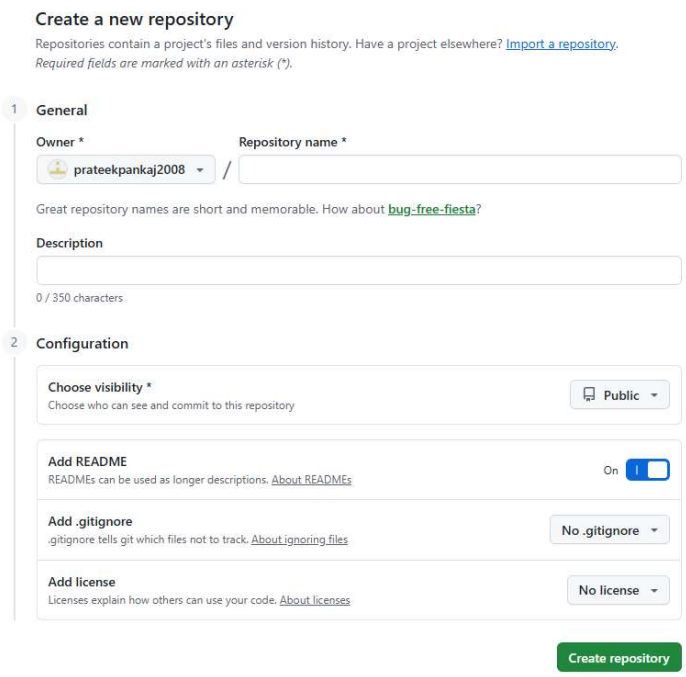
- **Git Commit –m "Write proper message"**

MINGW64:/c/Users/admin/Downloads/GIT_Tutorials

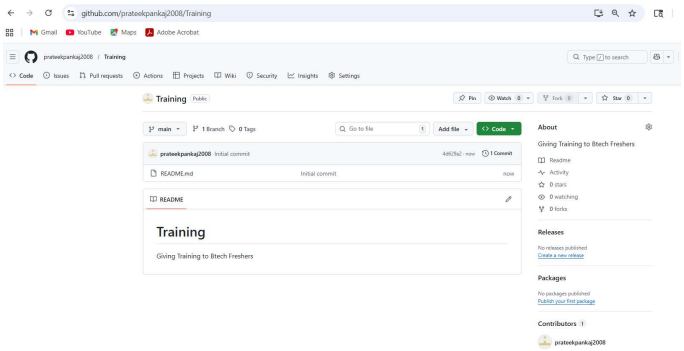
```
admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/GIT_Tutorials (master)
$ git add --a

admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/GIT_Tutorials (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        New Microsoft Word Document.docx
        pp.txt.txt

nothing added to commit but untracked files present (use "git add" to track)

admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/GIT_Tutorials (master)
$ git add *

admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/GIT_Tutorials (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   New Microsoft Word Document.docx
        new file:   pp.txt.txt


admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/GIT_Tutorials (master)
$
```

GIT HUB UI

(1) https://github.com/
(2) Create a website using gmail
(3) Verify account
(4) Create a new repository



(5) Repo created



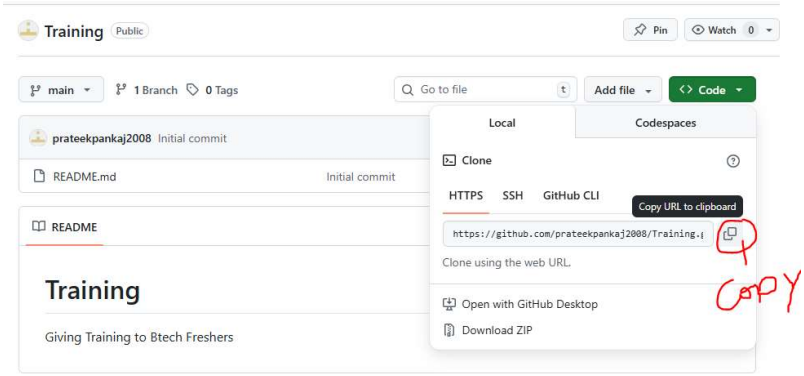
Link of repo is
https://github.com/prateekpankaj2008/Training

Now I want to add all python notes in repo

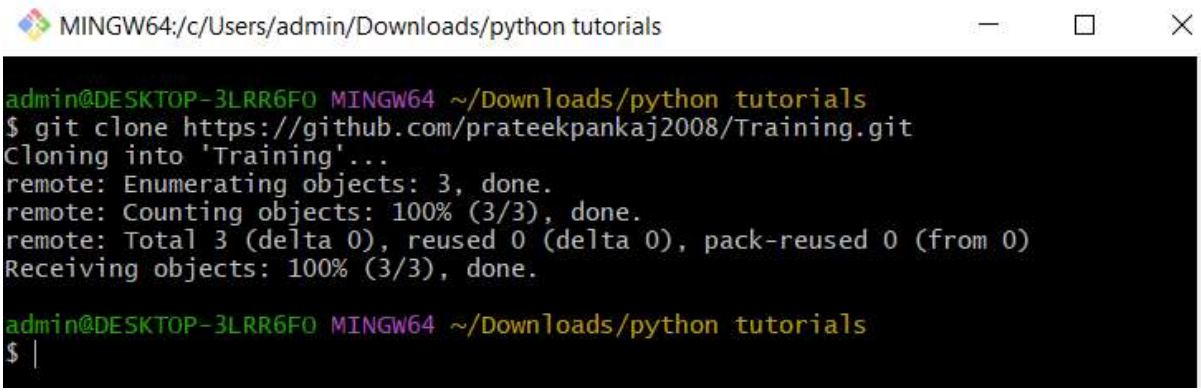
(1) Create a folder
C:\Users\admin\Downloads\python tutorials

(2) Right click and git bash
(3) Command prompt will open
(4) Go to git hub UI
https://github.com/prateekpankaj2008/Training
Take yours git hub link

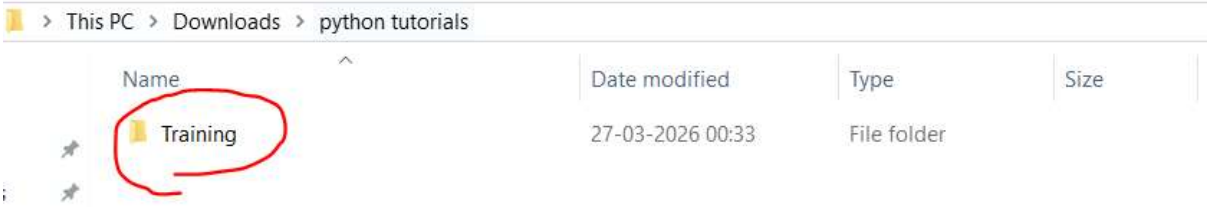
(5) Copy url,
Go to code which is in Green colour. Click on it.
And copy url



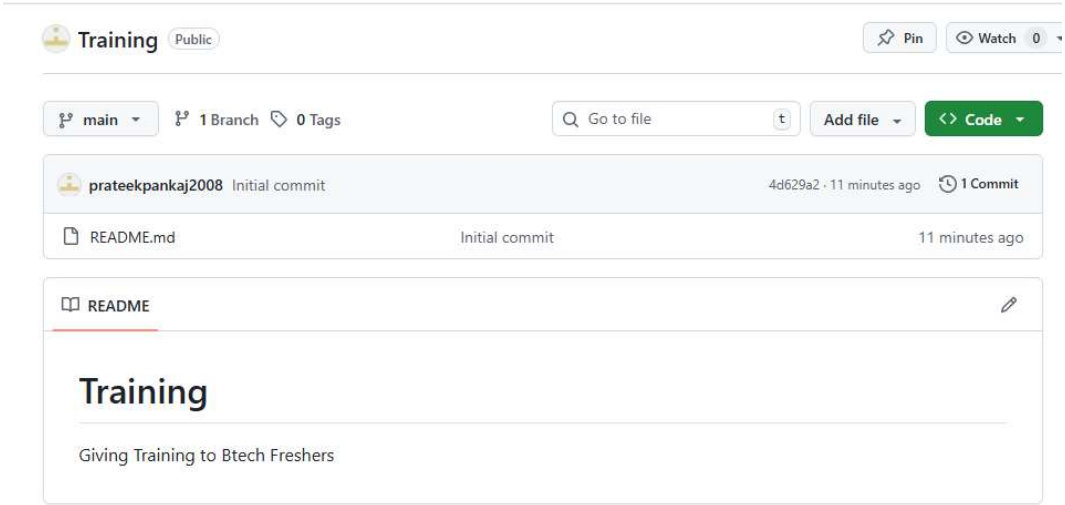
(6) Go to git bash command prompt and type
git clone url



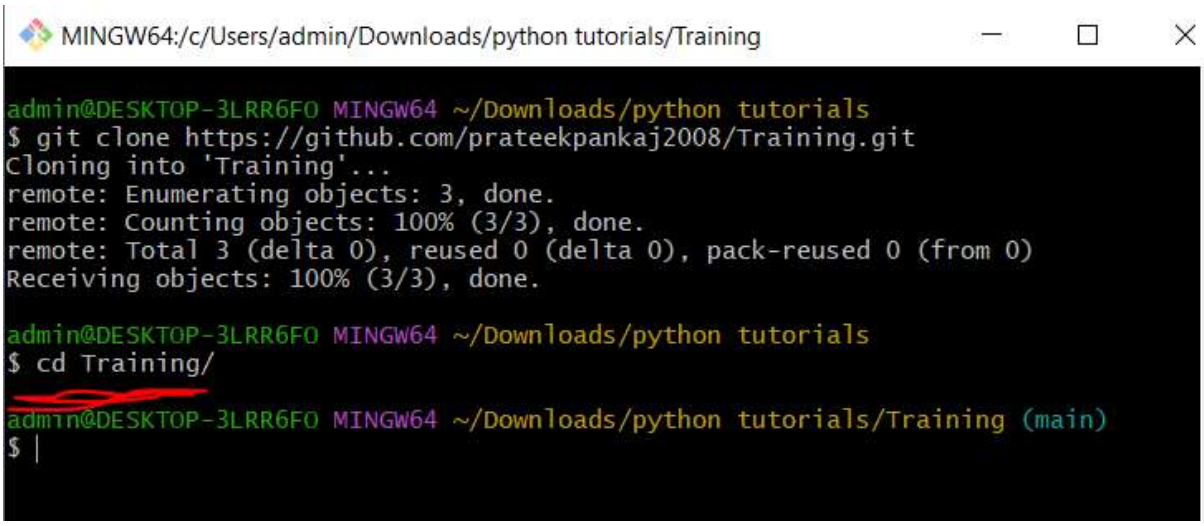
(7) Go to your folder and check what happened?



A new folder appeared which is same as your git hub UI repository name.

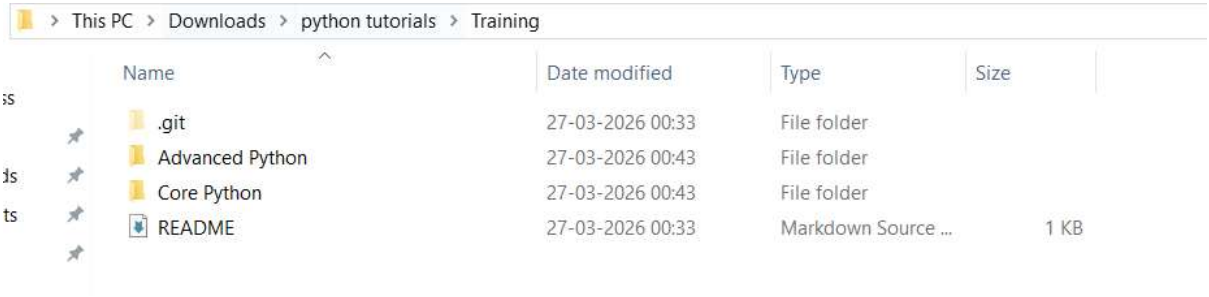


(8) Go to gitbash command prompt and go to folder Training



(9) Go to your folder , inside training, keep new data

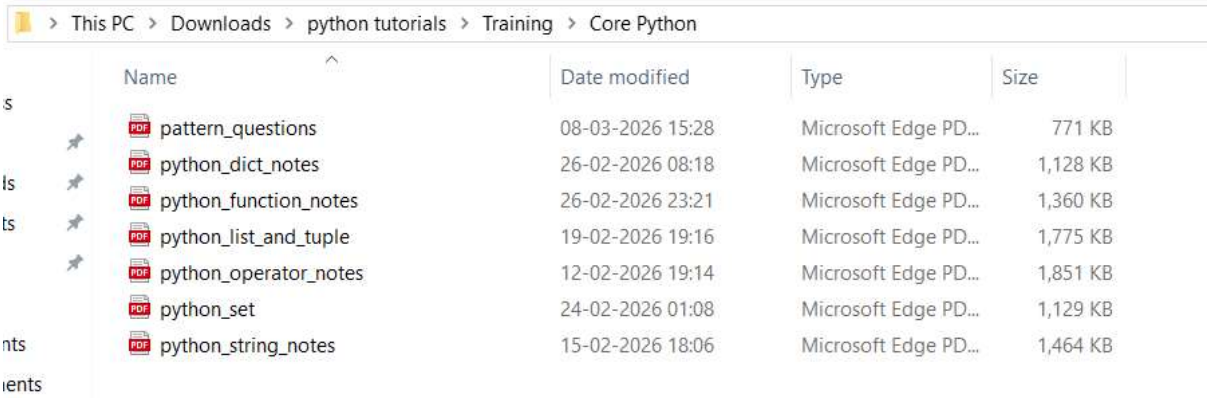
Added two folder



This PC > Downloads > python tutorials > Training

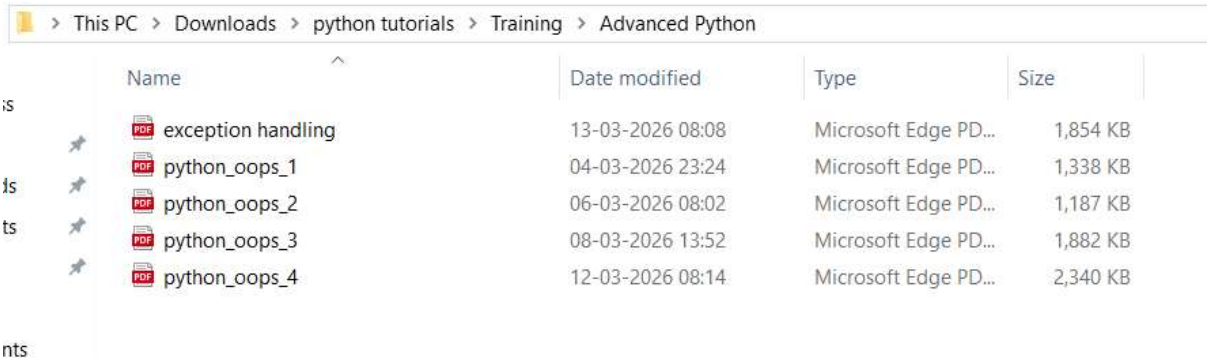
Name	Date modified	Type	Size
.git	27-03-2026 00:33	File folder	
Advanced Python	27-03-2026 00:43	File folder	
Core Python	27-03-2026 00:43	File folder	
README	27-03-2026 00:33	Markdown Source ...	1 KB

Inside folder added pdf files which are notes



This PC > Downloads > python tutorials > Training > Core Python

Name	Date modified	Type	Size
pattern_questions	08-03-2026 15:28	Microsoft Edge PD...	771 KB
python_dict_notes	26-02-2026 08:18	Microsoft Edge PD...	1,128 KB
python_function_notes	26-02-2026 23:21	Microsoft Edge PD...	1,360 KB
python_list_and_tuple	19-02-2026 19:16	Microsoft Edge PD...	1,775 KB
python_operator_notes	12-02-2026 19:14	Microsoft Edge PD...	1,851 KB
python_set	24-02-2026 01:08	Microsoft Edge PD...	1,129 KB
python_string_notes	15-02-2026 18:06	Microsoft Edge PD...	1,464 KB



This PC > Downloads > python tutorials > Training > Advanced Python

Name	Date modified	Type	Size
exception handling	13-03-2026 08:08	Microsoft Edge PD...	1,854 KB
python_oops_1	04-03-2026 23:24	Microsoft Edge PD...	1,338 KB
python_oops_2	06-03-2026 08:02	Microsoft Edge PD...	1,187 KB
python_oops_3	08-03-2026 13:52	Microsoft Edge PD...	1,882 KB
python_oops_4	12-03-2026 08:14	Microsoft Edge PD...	2,340 KB

(10) Go to gitbash command prompt and type commands

(a) git status

it will show non committed data, in red color

```
admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/python tutorials
$ cd Training/

admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/python tutorials/Training (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Advanced Python/
        Core Python/

nothing added to commit but untracked files present (use "git add" to track)

admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/python tutorials/Training (main)
$ |
```

(b) Type → git add *

Save all non committed data

© Again type → git status

```
MINGW64:/c/Users/admin/Downloads/python tutorials/Training
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Advanced Python/
        Core Python/

nothing added to commit but untracked files present (use "git add" to track)

admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/python tutorials/Training (main)
$ git add *

admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/python tutorials/Training (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   Advanced Python/exception handling.pdf
        new file:   Advanced Python/python_oops_1.pdf
        new file:   Advanced Python/python_oops_2.pdf
        new file:   Advanced Python/python_oops_3.pdf
        new file:   Advanced Python/python_oops_4.pdf
        new file:   Core Python/pattern_questions.pdf
        new file:   Core Python/python_dict_notes.pdf
        new file:   Core Python/python_function_notes.pdf
        new file:   Core Python/python_list_and_tuple.pdf
        new file:   Core Python/python_operator_notes.pdf
        new file:   Core Python/python_set.pdf
        new file:   Core Python/python_string_notes.pdf


admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/python tutorials/Training (main)
$ |
```

All data will be in green

(c) Commit all data

git commit –m “message”

git commit –m “Python notes added”

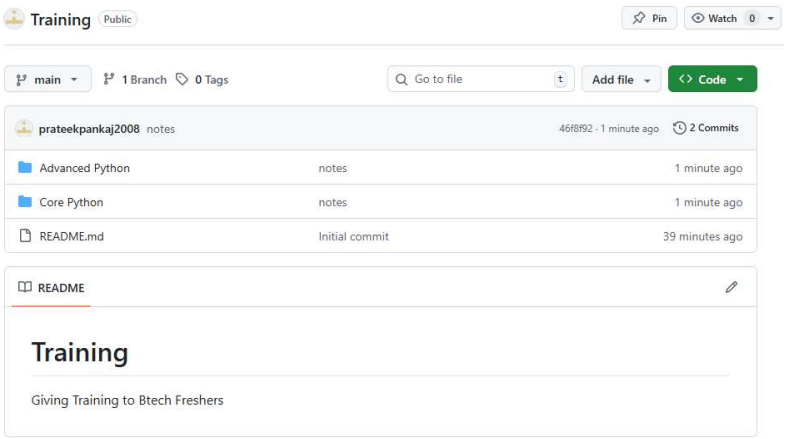
```
admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/python tutorials/Training (main)
$ git commit -m "python notes added"
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'admin@DESKTOP-3LRR6FO.(none)')

admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/python tutorials/Training (main)
$ |
```

Finally Push the code in Git repository

Using git push command

```
        new file:   Core Python/python_function_notes.pdf
        new file:   Core Python/python_list_and_tuple.pdf
        new file:   Core Python/python_operator_notes.pdf
        new file:   Core Python/python_set.pdf
        new file:   Core Python/python_string_notes.pdf


admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/python tutorials/Training (main)
$ git commit -m "notes"
[main 46f8f92] notes
 12 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Advanced Python/exception handling.pdf
 create mode 100644 Advanced Python/python_oops_1.pdf
 create mode 100644 Advanced Python/python_oops_2.pdf
 create mode 100644 Advanced Python/python_oops_3.pdf
 create mode 100644 Advanced Python/python_oops_4.pdf
 create mode 100644 Core Python/pattern_questions.pdf
 create mode 100644 Core Python/python_dict_notes.pdf
 create mode 100644 Core Python/python_function_notes.pdf
 create mode 100644 Core Python/python_list_and_tuple.pdf
 create mode 100644 Core Python/python_operator_notes.pdf
 create mode 100644 Core Python/python_set.pdf
 create mode 100644 Core Python/python_string_notes.pdf

admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/python tutorials/Training (main)
$ git push
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 4 threads
Compressing objects: 100% (16/16), done.
Writing objects: 100% (16/16), 15.19 MiB | 462.00 KiB/s, done.
Total 16 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/prateekpankaj2008/Training.git
   4d629a2..46f8f92  main -> main

admin@DESKTOP-3LRR6FO MINGW64 ~/Downloads/python tutorials/Training (main)
$ |
```

Go and check GITHUB UI



https://github.com/prateekpankaj2008/Training

SORTING – Selection Sort

Ascending order

```
l= [12, 45, 23, 67, 89, 33]

n = len(l)

for i in range(n):
    for j in range(i+1, n):
        if l[i]>l[j]:
            l[i], l[j] = l[j], l[i]

print(l)
```

Time complexity -> n square.

output => [12, 23, 33, 45, 67, 89]

Descending order

```
l= [12, 45, 23, 67, 89, 33]

n = len(l)

for i in range(n):
    for j in range(i+1, n):
        if l[i]<l[j]:
            l[i], l[j] = l[j], l[i]

print(l)
```